# The sealed statements of LocalAlgebrasLean

These 30 statements are the ones that `lake comparator` checks. Each is followed by the name of its formal version in `verify/Challenge.lean`.

## Nets

Let $R$ be a preordered set of regions and $A$ a C*-algebra. A star subalgebra is a unital complex star subalgebra, and a star isomorphism is a unital complex $*$-isomorphism. A region index on $R$ consists of an orthogonality relation $O_1 \perp O_2$ on the regions.

**1.** Region indices and their isomorphisms. A relation $r$ on $R$ is the orthogonality relation of a region index exactly when it is symmetric and antitone in its first argument ($O_0 \leq O_1$ and $r(O_1, O_2)$ give $r(O_0, O_2)$), and a region index is determined by its relation. For region indices $I$ on $R$ and $I'$ on a preordered set $R'$, with orthogonality relations $\perp$ and $\perp'$, an isomorphism from $I$ to $I'$ has a map $R \to R'$; a map $f$ is the map of an isomorphism exactly when it is the map of an order isomorphism and $f(O_1) \perp' f(O_2) \iff O_1 \perp O_2$ for all regions $O_1, O_2$, and an isomorphism is determined by its map. For a set $V$ of sites, the region index of the subsets of $V$, ordered by inclusion, has two subsets orthogonal exactly when they are disjoint.
`Sealed.region_index_iff`

**2.** Nets and generated nets. A net over $R$ in $A$ assigns to each region $O$ a star subalgebra $\mathcal{A}(O)$ of $A$; an assignment $O \mapsto F(O)$ of star subalgebras is a net exactly when it is monotone ($O \leq O'$ implies $F(O) \subseteq F(O')$, isotony) and every $F(O)$ is norm-closed, and a net is determined by its algebras. For a monotone family $O \mapsto G(O)$ of subsets of $A$, the generated net assigns to each region $O$ the norm closure of the star subalgebra of $A$ generated by $G(O)$ (a unital one, so that it contains the unit, and is $\mathbb{C}1$ when $G(O)$ is empty).
`Sealed.net_iff`

**3.** Properties of nets. Let $\mathcal{A}$ be a net over $R$ in $A$ and $I$ a region index on $R$ with orthogonality $\perp$. Generation: $\mathcal{A}$ is generating exactly when the norm closure of the star subalgebra generated by all its algebras is $A$. Local commutativity: $\mathcal{A}$ is locally commutative for $I$ exactly when, whenever $O_1 \perp O_2$, every element of $\mathcal{A}(O_1)$ commutes with every element of $\mathcal{A}(O_2)$. Covariance: let $\mathcal{A}'$ be a net over a preordered set $R'$ in a C*-algebra $A'$, $I'$ a region index on $R'$, $\kappa$ an isomorphism from $I$ to $I'$ and $\alpha : A \to A'$ a star isomorphism; then $(\kappa, \alpha)$ is a covariance pair of $\mathcal{A}$ and $\mathcal{A}'$ exactly when $\alpha(\mathcal{A}(O)) = \mathcal{A}'(\kappa(O))$ for every region $O$.
`Sealed.net_properties_iff`

**4.** Nets generated by sets. Let $O \mapsto G(O)$ be a monotone family of subsets of $A$ over $R$, with generated net $\mathcal{A}$. If the norm closure of the star subalgebra generated by the union of all $G(O)$ is $A$, then $\mathcal{A}$ is generating. Let $I$ be a region index on $R$ with orthogonality $\perp$: if every $G(O)$ is closed under the adjoint and every element of $G(O_1)$ commutes with every element of $G(O_2)$ whenever $O_1 \perp O_2$, then $\mathcal{A}$ is locally commutative for $I$. Let $I'$ be a region index on a preordered set $R'$, $A'$ a C*-algebra, $O \mapsto G'(O)$ a monotone family of subsets of $A'$ over $R'$ with generated net

$\mathcal{A}'$, $\kappa$ an isomorphism from $I$ to $I'$ and $\alpha : A \to A'$ a star isomorphism with $\alpha(G(O)) = G'(\kappa(O))$ for every region $O$: then $(\kappa, \alpha)$ is a covariance pair of $\mathcal{A}$ and $\mathcal{A}'$.
`Sealed.generated_net_properties`

## The qudit net

Let $\Lambda$ be a finite set of sites, each site $u$ carrying a finite alphabet $\sigma_u$; the alphabets may differ from site to site. A configuration $s$ chooses a letter $s(u) \in \sigma_u$ at every site. The qudit algebra is the algebra of complex matrices indexed by the configurations, with the matrix product and the conjugate transpose, a C*-algebra with the operator norm of the matrix acting on the Euclidean space of complex functions on the configurations. For a set of sites $X$, a matrix on the letters of $X$ is a complex matrix indexed by the assignments of a letter in $\sigma_x$ to each $x \in X$, and $s|_X$ is the restriction of a configuration $s$ to $X$.

**5.** The qudit algebra, the embedding and the qudit net. A configuration is a function assigning to each site $u$ a letter in $\sigma_u$, and the qudit algebra consists of the complex matrices indexed by the configurations. For a set of sites $X$ and a matrix $m$ on the letters of $X$, the embedding $j_X(m)$ is the matrix with

$$j_X(m)_{ss'} = m_{s|_X, s'|_X}$$

if the configurations $s$ and $s'$ agree at every site outside $X$, and 0 otherwise. The qudit net is a net over the sets of sites in the qudit algebra; a matrix $M$ lies in its algebra $\mathcal{A}(X)$ of a set of sites $X$ exactly when (1) $M_{ss'} = 0$ whenever $s$ and $s'$ differ at some site outside $X$, and (2) $M_{ss'} = M_{rr'}$ whenever $s$, $s'$ agree outside $X$, $r$, $r'$ agree outside $X$, $s = r$ on $X$ and $s' = r'$ on $X$.
`Sealed.qudit_net`

**6.** If every alphabet is nonempty, then for every set of sites $X$ the algebra $\mathcal{A}(X)$ of the qudit net consists exactly of the matrices $j_X(m)$ with $m$ a matrix on the letters of $X$. The algebra $\mathcal{A}(\Lambda)$ of the set of all sites is the whole qudit algebra. Local commutativity: the qudit net is locally commutative for the region index of the subsets of $\Lambda$, in which two sets of sites are orthogonal exactly when they are disjoint.
`Sealed.qudit_net_range_and_locality`

**7.** Relabelling covariance. Let $\Lambda$ and $\Lambda'$ be finite sets of sites with one finite alphabet $Q$ at every site, and $e : \Lambda \to \Lambda'$ a bijection. The relabelling by $e$ is a star isomorphism $\rho_e$ from the qudit algebra of $\Lambda$ onto the qudit algebra of $\Lambda'$ with

$$(\rho_e M)_{tt'} = M_{t \circ e, t' \circ e}$$

for every matrix $M$ in the qudit algebra of $\Lambda$ and all configurations $t, t'$ of $\Lambda'$; and $\rho_e(\mathcal{A}(X)) = \mathcal{A}'(e(X))$ for every set of sites $X \subseteq \Lambda$, where $\mathcal{A}$ and $\mathcal{A}'$ are the qudit nets of $\Lambda$ and $\Lambda'$.
`Sealed.relabel_covariance`

## Setting

There are sites $\Lambda$ (any set, possibly infinite), a finite alphabet $\sigma_u$ at every site $u$, and a reference configuration $s_0$ choosing a letter at every site (so every alphabet is nonempty). For a finite set of sites $X$, a matrix on the letters of $X$ is a complex matrix indexed by the assignments $\xi$ of a letter $\xi_x \in \sigma_x$ to each $x \in X$, that is, an operator on $\bigotimes_{x \in X} \mathbb{C}^{\sigma_x}$. Bounded operators carry the operator norm, and C*-algebras are complex.

**8.** The finitely supported configurations are the configurations $s$ that differ from the reference configuration $s_0$ at finitely many sites.
`Sealed.finConfig_def`

**9.** The Hilbert space $H$ is $\ell^2(F)$, the square-summable complex functions on the set $F$ of finitely supported configurations.
`Sealed.configSpace_def`

## The local embeddings

For a finite set of sites $X$, the local embedding $\iota_X$ is a unital $*$-homomorphism from the matrices on the letters of $X$; this is part of the type of $\iota_X$ in the Lean statements. Statement 10 determines $\iota_X$ by its matrix elements.

**10.** For a finite set of sites $X$, a matrix $m$ on the letters of $X$ and finitely supported configurations $s, t$, the matrix element $\langle e_s, \iota_X(m) e_t \rangle$ in the standard basis of $\ell^2(F)$ is $m(s|_X, t|_X)$ if $s$ and $t$ agree at every site outside $X$, and 0 otherwise.
`Sealed.inner_localEmbed`

**11.** For a finite set of sites $X$, the local embedding $\iota_X$ is injective and $\|\iota_X(m)\| = \|m\|$ for every matrix $m$ on the letters of $X$, the norm of $m$ being its operator norm on $\ell^2$.
`Sealed.localEmbed_injective_and_norm`

**12.** Compatibility: for finite sets of sites $X \subseteq Y$ and a matrix $m$ on the letters of $X$, $\iota_Y(m \otimes 1) = \iota_X(m)$, where

$$(m \otimes 1)(\zeta, \zeta') = m(\zeta|_X, \zeta'|_X)$$

if the letters $\zeta, \zeta'$ on $Y$ agree outside $X$, and 0 otherwise.
`Sealed.localEmbed_extend`

**13.** Local commutativity: for disjoint finite sets of sites $X$ and $Y$, $\iota_X(a)$ and $\iota_Y(b)$ commute for all matrices $a$ on the letters of $X$ and $b$ on the letters of $Y$.
`Sealed.commute_localEmbed_of_disjoint`

## The net of local algebras

Every set of sites $O$ has a $*$-subalgebra $\mathcal{A}(O)$ of the bounded operators on $H$, closed in norm (statement 14), and the quasi-local algebra is $\mathcal{A}_{ql} = \mathcal{A}(\Lambda)$.

**14.** For every set of sites $O$, the algebra $\mathcal{A}(O)$ is the norm closure, in the bounded operators on $H$, of the set of operators $\iota_X(m)$ with $X \subseteq O$ a finite set of sites and $m$ a matrix on the letters of $X$.
`Sealed.quasiLocalNet_eq_closure`

**15.** The quasi-local algebra $\mathcal{A}_{ql}$ is the algebra $\mathcal{A}(\Lambda)$ of the set of all sites.
`Sealed.quasiLocalAlgebra_def`

**16.** Local commutativity: for disjoint sets of sites $O_1$ and $O_2$, every element of $\mathcal{A}(O_1)$ commutes with every element of $\mathcal{A}(O_2)$.
`Sealed.commute_of_mem_quasiLocalNet`

**17.** Isotony: for sets of sites $O_1 \subseteq O_2$, $\mathcal{A}(O_1) \subseteq \mathcal{A}(O_2)$.
`Sealed.quasiLocalNet_mono`

## Symmetries and the reference configuration

**18.** Covariance: let a group $\Gamma$ act on the sites, with one finite alphabet $Q$ at every site and a reference configuration $s_0$ with $s_0(gu) = s_0(u)$. For each $g \in \Gamma$ there is a unitary $U$ on $H$ with $U\iota_X(m)U^* = \iota_{gX}(g_*m)$ for every finite set of sites $X$, where $gX = \{y : g^{-1}y \in X\}$ and

$$(g_*m)(\zeta, \zeta') = m(\zeta \circ g, \zeta' \circ g),$$

and with $U\mathcal{A}(O)U^* = \mathcal{A}(gO)$ for every set of sites $O$. The unitary is asserted for each $g$ separately; the statement does not say that $g \mapsto U$ is a representation.
`Sealed.exists_unitary_covariance`

**19.** Independence of the reference configuration: for any two reference configurations $s_0$ and $s_0'$ (which may differ at infinitely many sites) there is a $*$-isomorphism $\varphi : \mathcal{A}_{ql} \to \mathcal{A}'_{ql}$ of the quasi-local algebras with $a \in \mathcal{A}(O) \iff \varphi(a) \in \mathcal{A}'(O)$ for every set of sites $O$ and every $a \in \mathcal{A}_{ql}$, and with $\varphi(\iota_X(m)) = \iota'_X(m)$ for every finite set of sites $X$ and matrix $m$ on its letters.
`Sealed.exists_starAlgEquiv_of_ref`

## The universal property

**20.** Universal property: let $D$ be a C*-algebra and $(\beta_X)$ a family, indexed by the finite sets of sites $X$, of unital $*$-homomorphisms from the matrices on the letters of $X$ to $D$, compatible in the sense $\beta_Y(m \otimes 1) = \beta_X(m)$ for $X \subseteq Y$ (with $m \otimes 1$ as in the compatibility of the local embeddings). Then there is exactly one unital $*$-homomorphism $\Phi : \mathcal{A}_{ql} \to D$ with $\Phi(\iota_X(m)) = \beta_X(m)$ for all $X$ and $m$.
`Sealed.exists_unique_lift`

**21.** Let $D$ be a C*-algebra other than the zero algebra and $(\beta_X)$ a compatible family as in the universal property. Then the $*$-homomorphism $\Phi : \mathcal{A}_{ql} \to D$ with $\Phi(\iota_X(m)) = \beta_X(m)$ for all $X$ and $m$ is an isometry.
`Sealed.isometry_lift`

## Finite volume

**22.** If the set of sites $\Lambda$ is finite, the quasi-local algebra $\mathcal{A}_{ql}$ is the algebra of all bounded operators on $H$.
`Sealed.quasiLocalAlgebra_eq_top_of_finite`

**23.** For a finite set of sites $O$, the algebra $\mathcal{A}(O)$ is the range of the local embedding $\iota_O$.
`Sealed.quasiLocalNet_eq_range_of_finite`

## Examples

These examples show that the hypotheses of the main theorems can be met and that the objects are not degenerate.

**24.** For the chain $\mathbb{N}$ with two letters per site and the reference configuration 0, the quasi-local algebra is not the zero algebra.
`Sealed.witness_nontrivial_quasiLocalAlgebra`

**25.** For the chain $\mathbb{N}$ with two letters per site and the reference configuration 0, the set of finitely supported configurations, which indexes the basis of $H$, is infinite.
`Sealed.witness_infinite_finConfig`

**26.** For the chain $\mathbb{N}$ with two letters per site and the reference configuration 0, the algebra $\mathcal{A}(\{0\})$ of the site 0 contains an element that is not a multiple of the identity.
`Sealed.witness_quasiLocalNet_singleton_not_scalar`

**27.** For the chain $\mathbb{Z}$ with two letters per site and the reference configuration 0, which is invariant under translations, there is a unitary $U$ on $H$ with $U\mathcal{A}(\{0\})U^* = \mathcal{A}(\{1\})$.
`Sealed.witness_translation_covariant`

**28.** For the chain $\mathbb{N}$ with two letters per site, the reference configurations 0 and 1 differ at infinitely many sites, and there is a $*$-isomorphism $\varphi$ of their quasi-local algebras with $a \in \mathcal{A}(O) \iff \varphi(a) \in \mathcal{A}'(O)$ for every set of sites $O$.
`Sealed.witness_references_differing_everywhere`

**29.** For two sites $0, 1$ with two letters each, there is a monotone family $O \mapsto G(O)$, over the sets of sites, of subsets of the qudit algebra such that every $G(O)$ is closed under the adjoint, every element of $G(O_1)$ commutes with every element of $G(O_2)$ whenever $O_1$ and $O_2$ are orthogonal for the region index of the sets of sites, the norm closure of the star subalgebra generated by the union of all $G(O)$ is the whole qudit algebra, and $G(\{0\})$ contains two elements that do not commute.
`Sealed.witness_generated_net_hypotheses`

**30.** For two sites 0 and 1 carrying 2 and 3 letters, each of the algebras $\mathcal{A}(\{0\})$ and $\mathcal{A}(\{1\})$ of the qudit net contains two elements that do not commute.
`Sealed.witness_qudit_net_noncommutative`

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